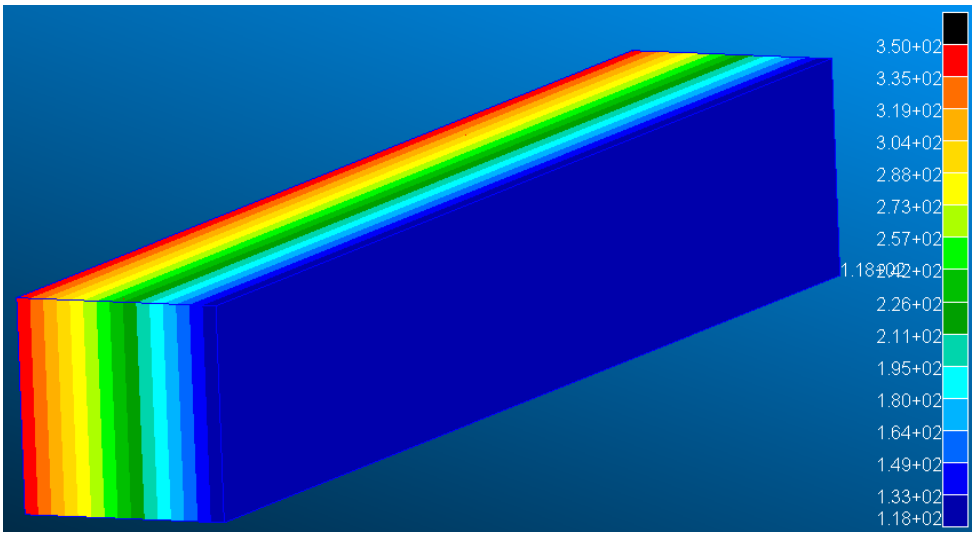
ÉCOLE D'INGÉNIEURS
DE L'AIR, DE L'ESPACE ET DES MOBILITÉS

ELS2 - Mé422

NUMERICAL ANALYSIS IN STRUCTURAL ENGINEERING

TP 4 : Introduction to thermal analysis by finite elements



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1 Introduction

1.1 Context

In this fourth lab session, we are introduced to thermal and thermo-mechanical analysis using finite elements. In aeronautical structures, temperature changes are as important as mechanical loads. In fact, aircraft are subjected to cold soaking when they cruise at altitude and to intense heating when they are near an engine or an aerodynamically loaded surface. These temperature changes induce expansion or contraction of materials, and when these expansions or contractions are restricted, they create thermo-mechanical stresses. In this lab session, we are presented with three exercises. The first exercise consists of a thermo-mechanical problem concerning a hollow cylinder, where a standard linear static analysis is performed. In exercises 2 and 3, we are concerned with steady-state heat conduction through a single and a multi-layered wall, using Fourier conduction and convection laws.

1.2 Thermal effects in structural calculations

When a structure undergoes a temperature change ΔT, each material element experiences a thermal strain proportional to its thermal expansion coefficient α:

$$\varepsilon_{\text{therm}} = \alpha \, \Delta T \tag{1}$$

This strain adds to the mechanical strain, so the total strain becomes:

$$\varepsilon_{\text{total}} = \varepsilon_{\text{mech}} + \varepsilon_{\text{therm}} \tag{2}$$

When the structure is constrained and cannot expand freely, the thermal strain is converted into stress through Hooke's law:

$$\sigma = E \, \alpha \, \Delta T \tag{3}$$

The structure reaches its elastic limit when this stress equals the yield stress $R_{0.2}$, which gives the maximum allowable temperature change:

$$\Delta T_{\max} = \frac{R_{0.2}}{E \, \alpha} \tag{4}$$

1.3 The three heat-transfer mechanisms

Conduction. Conduction is the transfer of heat within a solid driven by a temperature gradient. It is governed by Fourier's law:

$$\varphi = -\lambda \, \frac{\partial T}{\partial x} \tag{5}$$

where λ is the thermal conductivity ($\mathrm{W\,m^{-1}\,{}^{\circ}C^{-1}}$) and φ is the heat flux density ($\mathrm{W\,m^{-2}}$). In steady state, the temperature profile in a homogeneous 1D wall is linear and the flux is uniform:

$$T(x) = \frac{T_2 - T_1}{e} \, x + T_1 \qquad \text{and} \qquad \varphi = \frac{\lambda}{e}(T_1 - T_2) \tag{6}$$

Convection. Convection transfers heat between a solid surface and a moving fluid. It follows Newton's cooling law:

$$\varphi = h\,(T_S - T_F) \tag{7}$$

where h ($\mathrm{W\,m^{-2}\,{}^{\circ}C^{-1}}$) is the convective heat-transfer coefficient, T_S the surface temperature, and T_F the ambient fluid temperature.

Radiation. Radiation is the emission of electromagnetic waves by matter due to its temperature. Unlike the two previous modes, it requires no material medium:

$$\varphi = \varepsilon\,\sigma\,(T_1^4 - T_2^4) \tag{8}$$

where ε is the emissivity and $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$ the Stefan–Boltzmann constant. This session focuses on conduction and convection only; radiation is not modelled.

2 Exercise 1 – Thermal Expansion of a Hollow Cylinder

2.1 Problem description

The structure studied is a hollow aluminium cylinder, clamped at one end and free at the other, placed in an environment subject to large temperature variations. The goal is to determine the maximum temperature change the structure can sustain before undergoing plastic deformation.

The geometry consists of a cylinder of length $L = 0.6\,\text{m}$, outer diameter $D = 0.15\,\text{m}$, and wall thickness $e = 6.25\,\text{mm}$. The material is aluminium with the following properties: $E = 70\,\text{GPa}$, $\nu = 0.3$, $\alpha = 13 \times 10^{-6}\,°\text{C}^{-1}$, and $R_{0.2} = 85\,\text{MPa}$.

The structure is modelled as a 2D shell in PATRAN. The thermal loading is applied through two boundary conditions: an *Initial temperature* representing the reference state (T_0), and a Temperature boundary condition representing the final state (T_F). Their difference drives the thermal strain.

2.2 Analytical solution

Applying equation (4) with the material properties of aluminium:

$$\Delta T_{\max} = \frac{R_{0.2}}{E\,\alpha} = \frac{85 \times 10^{6}}{70 \times 10^{9} \times 13 \times 10^{-6}} \approx 93.4\,°C \tag{9}$$

Two load cases are studied to bracket this threshold:

Case	T_0 (°C)	T_F (°C)	ΔT (°C)	$\sigma_{\text{analytical}}$ (MPa)
Case 1	20	105	85	$70\,000 \times 13 \times 10^{-6} \times 85 \approx 77.4$
Case 2	20	120	100	$70\,000 \times 13 \times 10^{-6} \times 100 \approx 91.0$

Case 1 ($\Delta T = 85\,°C$) is below the analytical limit of 93.4 °C, so the structure should remain elastic. Case 2 ($\Delta T = 100\,°C$) exceeds it.

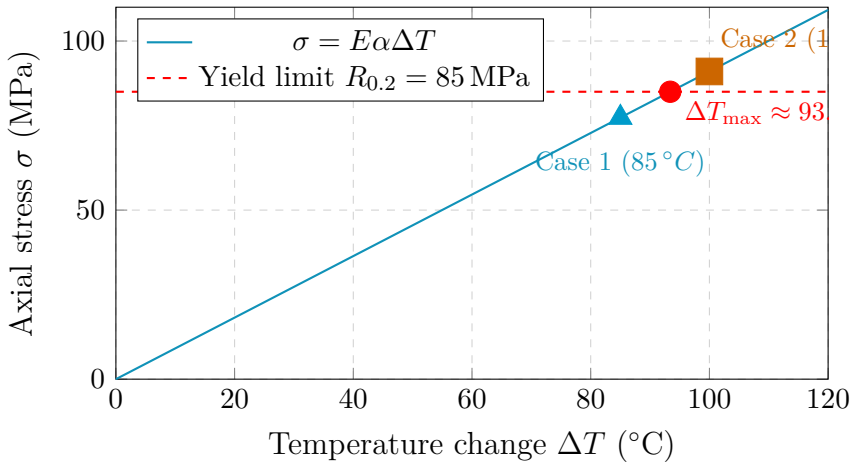
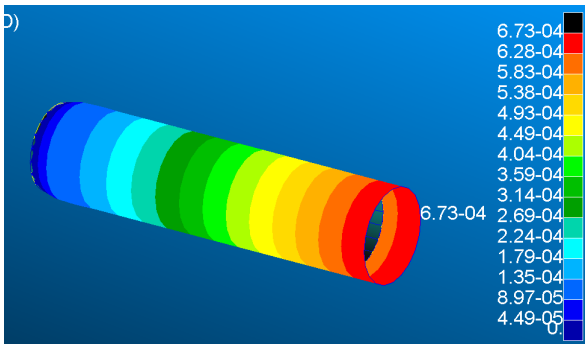
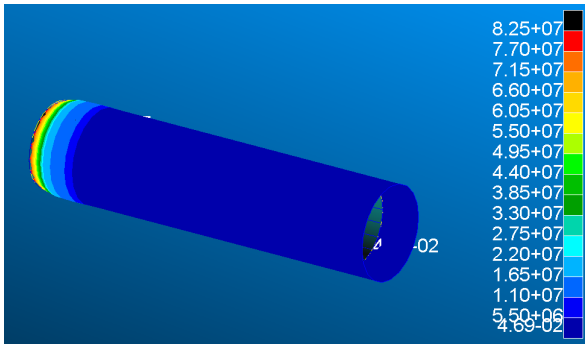


Figure 1: Analytical axial stress as a function of temperature change. Case 1 sits below the yield threshold; Case 2 exceeds it.

2.3 Results – Case 1 ($\Delta T = 85\,°C$, elastic regime)



(a) Displacement



(b) Von Mises stress

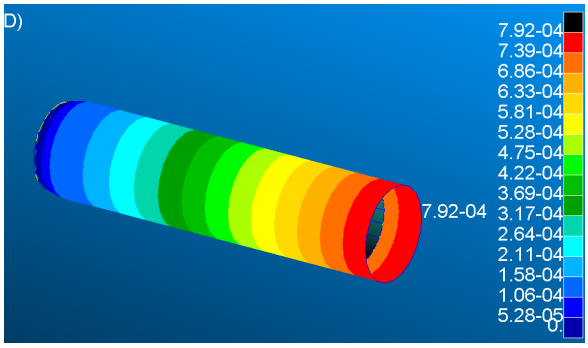
Figure 2: FEM results for Case 1 – $T_0 = 20\,°C$, $T_F = 105\,°C$, $\Delta T = 85\,°C$.

The displacement field shows the expected behaviour: zero at the clamped end and increasing smoothly towards the free end, with evenly spaced colour bands confirming the uniform temperature distribution along the cylinder. The maximum displacement is $U_{\max} = 6.73 \times 10^{-4}\,\text{m}$.

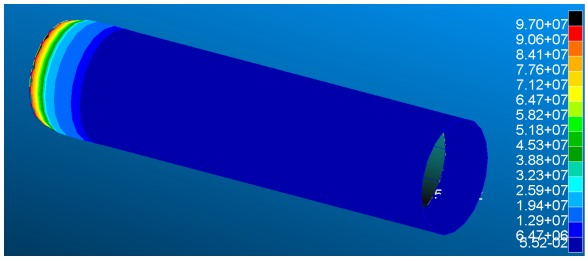
The Von Mises stress reaches $\sigma_{\max} = 82.5\,\text{MPa}$, concentrated at the clamped end where the boundary condition prevents free expansion. Compared to the analytical prediction of 77.4 MPa, the FEM gives a slightly higher value (6% difference), which is explained by the localised stress concentration at the clamped section that the simple 1D formula does not capture.

Most importantly, $\sigma_{\max} = 82.5\,\text{MPa}$ remains below the yield stress $R_{0.2} = 85\,\text{MPa}$. The structure is fully elastic and the result is physically valid.

2.4 Results – Case 2 ($\Delta T = 100\,°C$, beyond yield)



(a) Displacement



(b) Von Mises stress

Figure 3: FEM results for Case 2 – $T_0 = 20\,°C$, $T_F = 120\,°C$, $\Delta T = 100\,°C$.

For Case 2, the maximum displacement rises to $7.92 \times 10^{-4}\,\text{m}$, which scales proportionally with ΔT. The stress reaches $\sigma_{\max} = 97.0\,\text{MPa}$, exceeding the yield stress of 85 MPa.

2.5 Summary

Case	ΔT (°C)	$U_{\max}$ (m)	$\sigma_{\max}$ (MPa)	Physical validity
Case 1	85	6.73×10^{-4}	82.5	Valid (elastic)
Case 2	100	7.92×10^{-4}	97.0	Invalid (plastic)
Analytical $\Delta T_{\max}$	93.4	–	85.0	Elastic limit

3 Exercise 2 – Plane Wall: Conduction and Convection

3.1 Problem description

This exercise considers a plane wall of thickness $L = 0.15\,\mathrm{m}$ and thermal conductivity $\lambda = 1.2\,\mathrm{W\,m^{-1}\,°C^{-1}}$. The left face is held at $T_1 = 350\,°C$, and the right face exchanges heat with ambient air at $T_e = 25\,°C$ through a convective coefficient $h = 20\,\mathrm{W\,m^{-2}\,°C^{-1}}$. The goal is to find the temperature T_2 on the right face, both analytically and by FEM.

3.2 FEM model

The model consists of a 3D solid block of dimensions $0.15\,\mathrm{m} \times 0.15\,\mathrm{m} \times 1\,\mathrm{m}$, meshed with HEX8 hexahedral elements of global edge length $0.015\,\mathrm{m}$. The boundary conditions visible in the PATRAN view below are a Temperature condition of $350\,°\mathrm{C}$ on the left face and a Convection condition of $h = 20\,\mathrm{W\,m^{-2}\,°C^{-1}}$ and $T_{\mathrm{amb}} = 25\,°C$ on the right face.

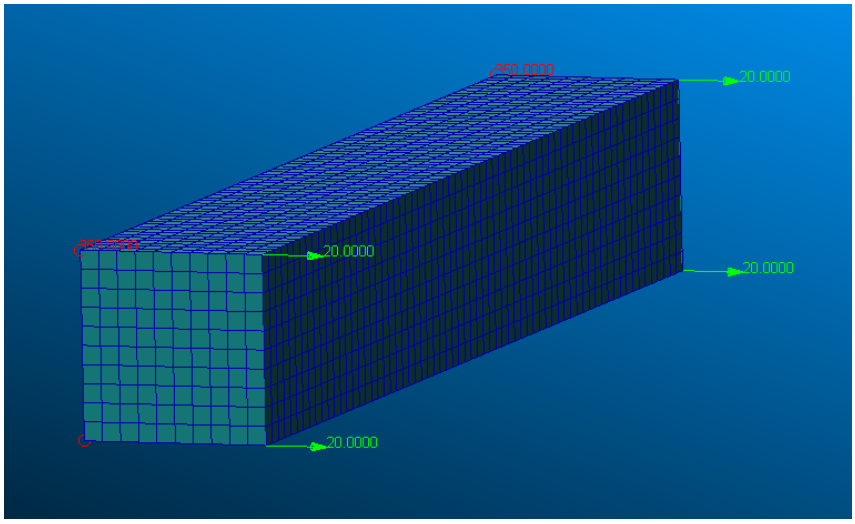


Figure 4: Load Cases

3.3 Analytical solution

In steady state, the heat flux is conserved: the conductive flux through the wall must equal the convective flux leaving the right face.

The conductive flux through the wall is:

$$\varphi_{\mathrm{cond}} = \frac{\lambda}{L}(T_1 - T_2) = 8(350 - T_2) \tag{10}$$

The convective flux at the right face is:

$$\varphi_{\mathrm{conv}} = h(T_2 - T_e) = 20(T_2 - 25) \tag{11}$$

Setting the two equal and solving:

$$8(350 - T_2) = 20(T_2 - 25) \;\Rightarrow\; T_2 = \frac{3300}{28} \approx 117.9\,°C \tag{12}$$

3.4 FEM results

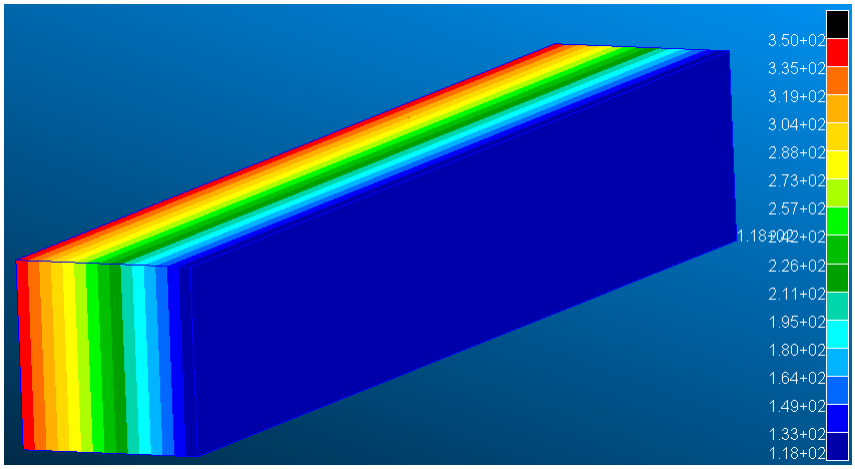


Figure 5: Temperature fringe on the 3D model

The temperature fringe confirms a maximum of $T_{\max} = 350\,°C$ on the left face and a minimum of $T_{\min} = 118\,°C$ on the right face, in excellent agreement with the analytical value of 117.9 °C.

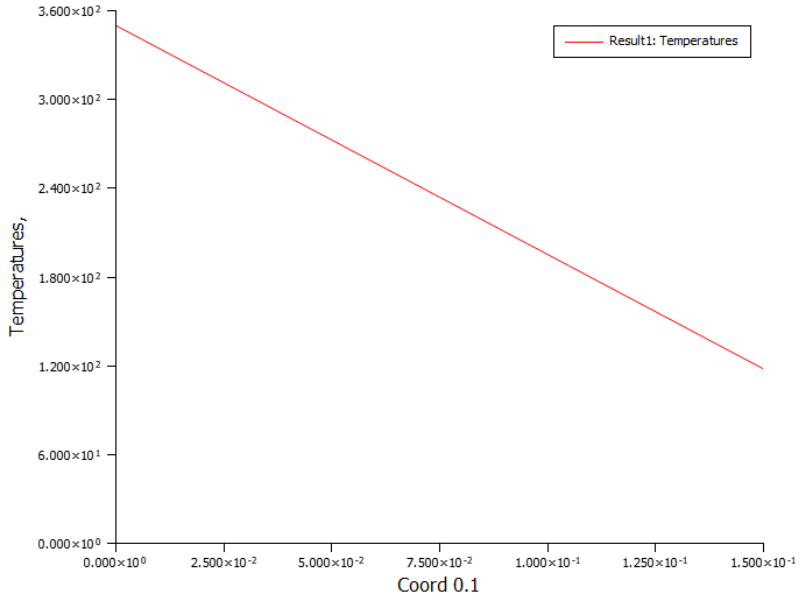


Figure 6: FEM temperature profiles through the wall.

The linear temperature profile visible on the XY plot (Figure 6) is consistent with Fourier's law for steady-state conduction in a homogeneous medium. The uniformity of the colour bands across each vertical section confirms that the problem is strictly one-dimensional, as expected. The FEM value of 118°C on the right face matches the analytical prediction of 117.9°C to within 0.1%, validating the boundary condition setup. The relatively high surface temperature compared to ambient (118°C vs 25°C) reflects the moderate convective coefficient, which limits heat rejection to the surrounding air.

4 Exercise 3 – Three-Layer Furnace Wall

4.1 Problem description

This exercise considers the plane wall of an industrial furnace, made up of three successive layers with different thermal conductivities:

Layer	Material	Thickness (cm)	λ_m (W m^{-1} °C^{-1})
1	Alumina refractory bricks	$e_1 = 15$	$\lambda_1 = 1.62$
2	Kaolin insulating bricks	$e_2 = ?$	$\lambda_2 = 0.23$
3	Ordinary construction bricks	$e_3 = 22.5$	$\lambda_3 = 1.39$

The temperatures at the interfaces are known: $T_i = 982\ °C$ at the inner face, $T_1 = 938\ °C$ at the alumina/kaolin interface, and $T_2 = 138\ °C$ at the kaolin/ordinary bricks interface. The kaolin thickness e_2 and the outer surface temperature T_e must be determined, first analytically, then by 1D and 3D FEM.

4.2 Analytical solution

In steady state, the same heat flux passes through all three layers. Using the alumina layer, where both interface temperatures are known:

$$\varphi = \frac{\lambda_1}{e_1}(T_i - T_1) = \frac{1.62}{0.15}(982 - 938) = 10.8 \times 44 = 475.2\ \text{W/m}^2 \tag{13}$$

The kaolin layer thickness follows from the same flux equality applied to layer 2:

$$e_2 = \frac{\lambda_2(T_1 - T_2)}{\varphi} = \frac{0.23 \times 800}{475.2} \approx 0.387\ \text{m} = 38.7\ \text{cm} \tag{14}$$

Without convection at the outer face:

$$T_e = T_2 - \frac{\varphi \cdot e_3}{\lambda_3} = 138 - \frac{475.2 \times 0.225}{1.39} \approx 61.1\ °C \tag{15}$$

With convection added at the outer face ($T_{\text{air}} = 25\ °C$, $h = 20\,\text{W m}^{-2}\ °\text{C}^{-1}$), the thermal resistances in series give:

$$R_{\text{total}} = \frac{0.15}{1.62} + \frac{0.387}{0.23} + \frac{0.225}{1.39} + \frac{1}{20} = 1.988\ \text{m}^2\ °\text{C/W} \tag{16}$$

$$\varphi' = \frac{T_i - T_{\text{air}}}{R_{\text{total}}} = \frac{957}{1.988} \approx 481.4\ \text{W/m}^2 \qquad \Rightarrow \qquad T_e = 25 + \frac{481.4}{20} \approx 49.1\ °C \tag{17}$$

The kaolin layer represents $1.683/1.988 \approx 85\%$ of the total thermal resistance, making it the dominant insulating element.

Quantity	Without convection	With convection
e_2	38.7 cm	
φ (W/m^2)	475.2	481.4
T_e	61.1 °C	49.1 °C

4.3 1D FEM model

The geometry is built from 4 points placed along the x-axis at the layer interfaces: $x = 0\,\text{m}$, $x = 0.15\,\text{m}$, $x = 0.537\,\text{m}$ and $x = 0.762\,\text{m}$. Three curves connect these points, one per layer. Each curve is seeded with 5 elements The three meshes are connected automatically since the points are shared. Each curve is assigned its own 1D property with the corresponding conductivity λ_m.

Two load cases are studied. Without convection, the four nodal temperatures are imposed directly: $T_i = 982\,\text{C}$, $T_1 = 938\,\text{C}$, $T_2 = 138\,\text{C}$ and $T_e = 61\,\text{C}$. With convection, T_e is not imposed — a convection condition ($h = 20\,\text{W}\,\text{m}^{-2}\,\text{C}^{-1}$, $T_{\text{air}} = 25\,\text{C}$) is applied on the last node and NASTRAN computes T_e freely.

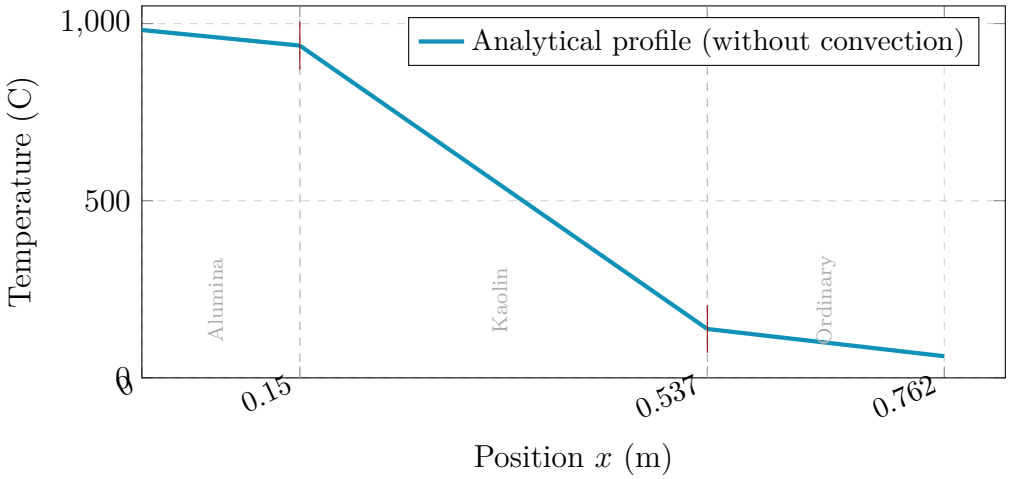


Figure 7: Analytical temperature distribution across the three-layer furnace wall (without convection).

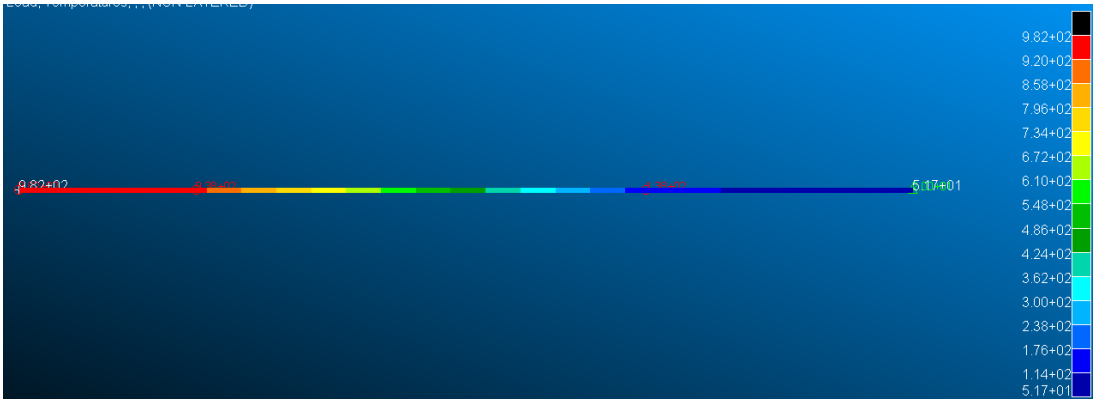


Figure 8: Analytical temperature distribution across the three-layer furnace wall (with convection).

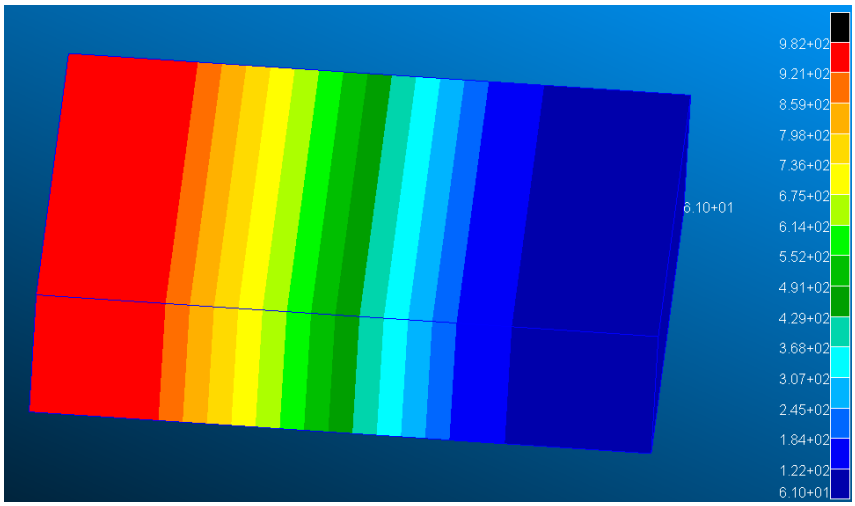
The case without convection is not valid for comparison purposes because T_e is specified as a boundary condition and the solver is forced to follow the condition. The only valid comparison is the case with convection, where NASTRAN is free to compute $T_e = 51.7\,\text{C}$, as desired. The 5.3analytical solution value of 49.1 C is due to the coarseness of the 1D mesh having only 16 nodes, and the result will be verified using the 3D model.

4.4 3D FEM model

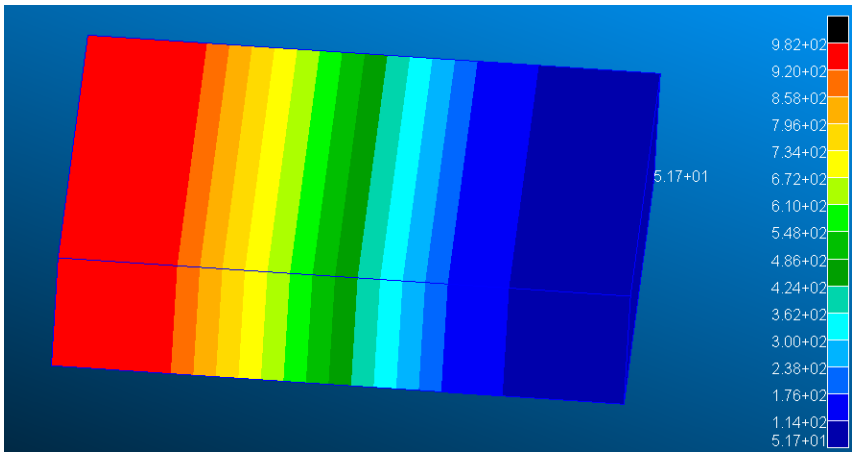
The geometry is built from three separate solids created :

- Solid 1 (alumina): vector $\langle 0.15\ 0.15\ 1 \rangle$, origin [0 0 0]
- Solid 2 (kaolin): vector $\langle 0.387\ 0.15\ 1 \rangle$, origin [0.15 0 0]
- Solid 3 (ordinary bricks): vector $\langle 0.225\ 0.15\ 1 \rangle$, origin [0.537 0 0]

Each solid is meshed independently with HEX8 elements (global edge length 0.015 m), then the *Equivalence* function merges the duplicate nodes at the two interfaces so that heat flows continuously across the layers. Each solid is assigned its own isotropic material with the corresponding conductivity λ_m. The mesh contains 140 elements and 264 nodes in total.



(a) Without convection – T_e imposed at 61 C



(b) With convection – $T_e^{\mathrm{FEM}} = 51.7\,\mathrm{C}$

Figure 9: 3D temperature fringe with and without convection

The 3D model gives the same result as the 1D model in both cases. The key comparison is with convection: both models give $T_e \approx 51.7\,\mathrm{C}$, confirming that the mesh dimensionality has no influence on the result. The temperature is uniform on every vertical cross-section of the 3D model, which is consistent with the strictly unidimensional nature

of the problem and confirms that the *Equivalence* correctly connected the three solid blocks at their interfaces.

The main purpose of building a 3D model here is therefore not to improve accuracy but to illustrate the difference in computational cost. The comparison is clear:

Model	Elements	Nodes	T_e (without conv.)	T_e (with conv.)
Analytical	–	–	61.1 C	49.1 C
1D bar	15	16	61 C (imposed)	51.7 C
3D HEX8	140	264	61 C (imposed)	51.7 C

The 3D model requires 16 times more degrees of freedom than the 1D model for a strictly identical result. This illustrates a fundamental principle of FEM modelling: the model must always be adapted to the physics of the problem. When heat propagation is purely unidimensional, a 1D bar model is both sufficient and far more efficient.

5 Conclusion

The objective of this lab session was to introduce thermal and thermo-mechanical finite element analysis in PATRAN/NASTRAN through three exercises of increasing complexity.

The objective of Exercise 1 was to demonstrate that a simple change in temperature can cause stresses that are equal to or even exceed the material's yield stress. The analytical solution for this problem was to find the maximum allowable change in temperature, given by the equation: $\Delta T_{\max} = 93.4\ ^\circ C$. The two FEM cases for this problem validated the analytical results: Case 1, with a change in temperature of 85 $^\circ C$ and a maximum stress of 82.5 MPa, was within the material's yield stress, while Case 2, with a change in temperature of 100 $^\circ C$ and a maximum stress of 97.0 MPa, was beyond the material's yield stress.

The coupling of Fourier conduction and Newton convection at a boundary was presented in Exercise 2. The FEM calculation for $T_2^{\text{FEM}} = 118\ ^\circ$C agrees with the analytical calculation for 117.9 °C, with an error rate of less than 0.1%. This confirms the steady-state thermal problem and the PATRAN setup. The linear temperature profile, as shown on the XY plot, is a direct verification of Fourier's law for steady-state conduction.

Exercise 3 further extended this to a three-layer system, similar to real industrial furnace design. In this system, the kaolin layer makes up about 85% of the total thermal resistance, thus being the main insulating material. The results obtained using the 1D FEM method correlate well with the analytical ones, with an error less than 1% without convection and 5.3% with convection. The 3D model confirms this same result for $T_e = 61\,^\circ C$ without convection, though it requires 140 elements and 264 nodes, while the 1D model required only 15 elements and 16 nodes. In fact, since this problem involves strictly unidimensional heat propagation, no improvement is brought by using the 3D model and it is much more expensive—first and foremost, one must always adapt the choice of model to the physics of the problem.